# MULTIPLE DISEASE PREDICTION SYSTEM

## INTERNSHIP REPORT

**Submitted to**

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**by**

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**Under the guidance of**

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**in partial fulfillment of the requirements for the award of the degree of**

**Bachelor of Engineering**

**Department of Computer Science and Engineering**

**SDM INSTITUTE OF TECHNOLOGY**

**UJIRE – 574 240**

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# 1. Executive Summary

This report refers to work completed during my internship with Techciti Software Consulting Pvt. Ltd at Bangalore from August 14th, 2023 until September 9th, 2023.

Artificial Intelligence and Machine Learning is the key area of my internship. In this internship I got the opportunity to develop a Multiple disease prediction System. The purpose of this project is to predict the chances of a disease to the person.

The learning objective of this internship is to gain knowledge on Artificial Intelligence and Ma

chine Learning. This work deals with the design and implementation of the algorithm that is required to build the system. The programing skill we learnt during internship are Python libraries and frameworks. Was able to individually develop this system which includes home page login page and prediction page.

On the whole, this internship was a useful experience. I have gained new knowledge, skills and met many new people. Skills I learned will give me the chance of future personal and professional growth. The result I had got was better than I ever expected. As a personal experience, this internship made my communication skills better and all around.

The main motive behind choosing Techciti Software Consulting Pvt. Ltd is that, they offered me a highly dynamic, fast-paced, professional working environment and chance to develop the skills I gained at university in a real business environment. It is one of the few companies that empower me an employee and helps to accelerate our career. The main purpose of this internship is to have an actual work experience in the field of study in order to test theoretically knowledge through practical work and apply this knowledge in a real-life situation. This internship helped me to bridge academic preparation in statistics with real world applied research or data science applications.

# 2. Company Profile

## 2.1 About Company

Since established in 2013, TechCiti has become a pioneer in providing distinguished end-to-end IT infrastructure solutions to its customers through our business functions maximizing customer engagement with personalized services. We believe that today more than ever, businesses are dependent on technology solutions.

The company have successfully established in their business functions across 12 + major cities across PAN India with over 56 + satisfied corporate customers.

**Website:** http://www.techciti.in/

**Specialties:** IT Security, Server, Databases, Networks, End User Services, Cloud Virtualization, Application development

## 2.2 Vision and Mission

**Vision: "**Our vision is to enable people and organizations realize their potential reinventing their engagement in defining the future using - technology."

**Mission:** "Our mission is to achieve the leading position as a distinguished & absolute end-to-end information technology infrastructure & service provider. We want to develop with profitable growth through superior Customer service, Innovation, Quality and Commitment".

## 2.3 Services of TechCiti Technologies

Services of Techciti are based on Web App Development, Android App Development, Digital Marketing, Machine Learning and Data Science, Advertising and motion graphics.


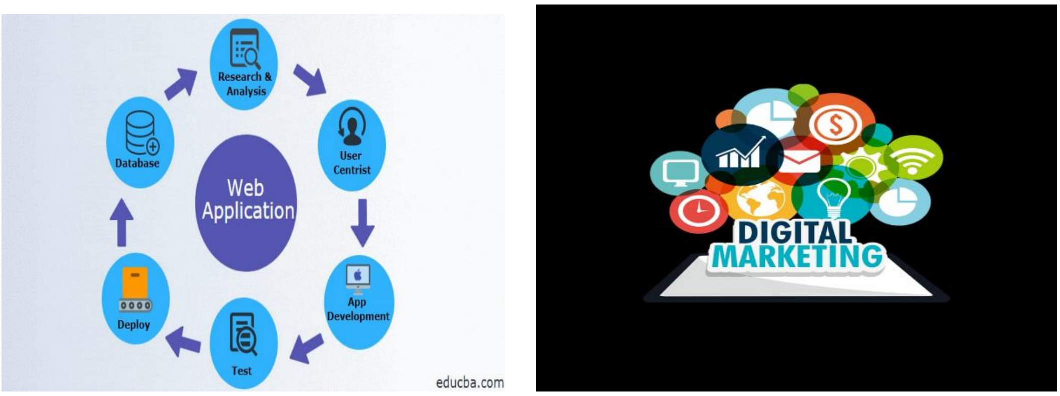


**Figure 2.1: Services of Techciti Software**

## 2.4 Clients of Techciti Technologies

Organizations need to develop new technology, to keep up with business requirements; that can help for reliably migrate physical and virtual servers, and data in real-time replication, from a single, intuitive user console. Some clients of Techciti software are as follows,

AURIGENE
PHARMACEUTICAL SERVICES

Delivering the Digital Future
Netcracker
An NEC Company

riverbed
The Digital Performance Company

**Figure 2.2: Clients of Techciti Technologies**

# 3.Problem Statement and Objectives

## 3.1 Problem Statement

Healthcare has always been a critical concern worldwide, and early detection of diseases can significantly improve patient outcomes and reduce healthcare costs. However, the traditional diagnostic methods often lack accuracy and efficiency, leading to delayed or incorrect diagnoses. Furthermore, the increasing prevalence of multiple diseases such as diabetes, Parkinson's disease and necessitates the development of an advanced prediction system that can accurately identify these diseases at an early stage. Currently, there is a lack of comprehensive systems capable of simultaneously predicting multiple diseases using patient data. Existing prediction models often focus on single diseases, limiting their effectiveness in providing a holistic health assessment. Additionally, the integration of multiple machine learning algorithms and techniques into a unified prediction system poses several challenges, including feature selection, data preprocessing, model selection, and performance evaluation.

## 3.2 Objectives

The main objectives are.

- To predict the chances of a disease to a person.

  By entering the data to the specified data fields user can come to know whether he/she is prone to the disease.
- To integrate the individual disease prediction models into a unified MDPS framework.

  To provide a single platform where a user can have a check for multiple diseases.
- To deploy the MDPS as a user-friendly and accessible tool for healthcare practitioners, enabling early disease detection and intervention to improve patient outcomes.

# 4.Weekly Overview of Internship

The internship was carried out for four weeks starting from 14th August to 9th September, 2023. The following tables provide the description of daily work done in four weeks.

**Table 4.1: Week 1 Work Done**

| | Date | Day | Task/ Topic Completed |
|---|---|---|---|
| **Week 1** | 14/08/2023 | Mon | Registration for Machine Learning Course |
| | 15/08/2023 | Tue | Introduction to basics of Front-End Design, such as HTML, CSS, Django, JavaScript and Databases |
| | 16/08/2023 | Wed | Installation of required software, Visual Studio Code Setup. Description about HTML tags |
| | 17/08/2023 | Thu | Creating HTML Login page |
| | 18/08/2023 | Fri | CSS properties, CSS Selectors |

**Table 4.2: Week 2 Work Done**

| | Date | Day | Task/ Topic Completed |
|---|---|---|---|
| **Week 2** | 21/08/2023 | Mon | Python basics, Data types and Functions |
| | 22/08/2023 | Tue | Python libraries: Introduction to matplotlib |
| | 23/08/2023 | Wed | Python libraries: Introduction to numpy<br>Python libraries: Introduction to sklearn |
| | 24/08/2023 | Thu | Introduction to TensorFlow for deep learning in Python |
| | 25/08/2023 | Fri | Introduction to Bootstrap, Front End Design Completions |

**Table 4.3: Week 3 Work Done**

| | Date | Day | Task/ Topic Completed |
|---|---|---|---|
| Week 3 | 28/08/2023 | Mon | Introduction to Machine Learning, uses history and overview of sample projects |
| | 29/08/2023 | Tue | Introduction to Machine Learning: Supervised learning Algorithms and unsupervised learning |
| | 30/08/2023 | Wed | Creating the front end of Multiple disease prediction system page using Flask |
| | 31/08/2023 | Thu | Project Implementation: Collection of Raw data, importing the necessary modules and filtering the data |
| | 01/09/2023 | Fri | Work was reviewed by supervisor and gave information regarding next procedure |

**Table 4.4: Week 4 Work Done**

| | Date | Day | Task/ Topic Completed |
|---|---|---|---|
| Week 4 | 04/09/2023 | Mon | Project Implementation: Started working on the backend functioning. |
| | 05/09/2023 | Tue | Project Implementation: Completed work on backend with few errors |
| | 06/09/2023 | Wed | Project guide reviewed the work and helped in clearing errors |
| | 07/09/2023 | Thu | Established connection between front end and back end |
| | 08/09/2023 | Fri | Final review by supervisor, collected all the feedbacks, made all necessary changes with their supervision |

# 5.Training Outline

## 5.1 Technologies Used

### Streamlit

Streamlit is an open-source Python library that simplifies the process of creating interactive web applications for data science and machine learning projects. It allows developers and data scientists to build intuitive and visually appealing apps with minimal effort, primarily using Python scripts.

### Scikit-learn

Scikit-learn is one of the most popular and widely used open-source machine learning libraries in Python. It is built on top of other scientific computing libraries such as NumPy, SciPy, and Matplotlib, and it provides a simple and efficient toolset for data mining and data analysis tasks.

### Pickle

Pickle is a Python module used for serializing and deserializing Python objects. Serialization is the process of converting an object into a byte stream, which can then be stored in a file or transmitted over a network. Deserialization is the process of reconstructing the original object from the byte stream.

## 5.2 Tools Used

### Visual studio

Visual Studio is an integrated development environment (IDE) developed by Microsoft. It provides a comprehensive set of tools and features for software development across various platforms, including Windows, macOS, and Linux. Visual Studio supports multiple programming languages such as C#, C++, Python, JavaScript, and more.

### Google Chrome Web Browsers

The Google Chrome Web browser is based on the open-source Chromium project. It is available for Windows, Mac OS and Linux operating systems. Each open website runs as its own process, which helps prevent malicious code on one page from affecting others.

## 5.3 Methodology Used

The methodology for developing a crop recommendation system typically involves several steps:

1. Input Data: Gather data of patients on various factors like blood pressure, insulin level etc.
2. Preprocessing: Clean the data by handling missing values, outliers and prepare the dataset.
3. Model Selection: Choose appropriate machine learning models for the task.
4. Model Configurations: To achieve higher test and cross validation accuracy, we use various configurations such as logistic regression, decision trees, random forests, support vector machines,
5. Training Models: Split the data into training and validation sets, and train the selected models on the training data.
6. Testing Accuracy of the Model: We evaluate the accuracy of the created model against the test data. In addition, we measured the cross-validation accuracy of the model. If the accuracy is unsatisfactory, we iterate the process by returning to the "Model Configuration" step.

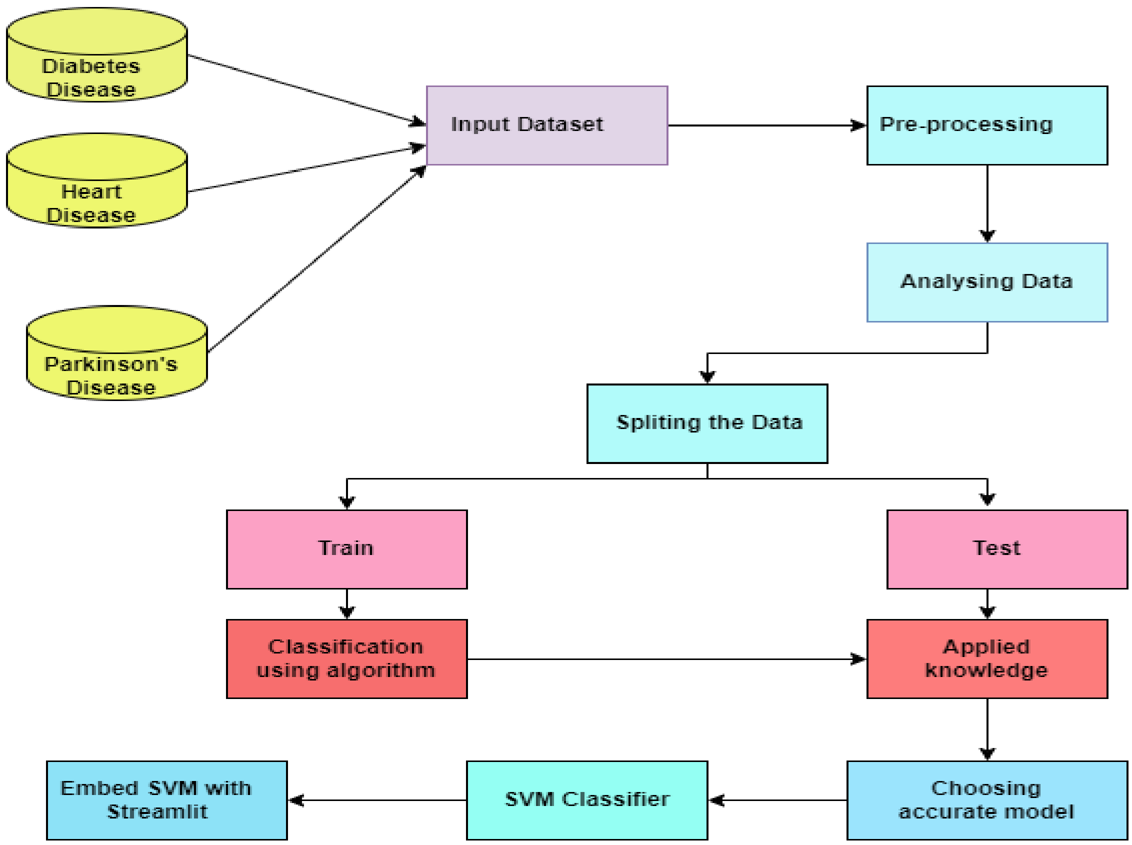


**Figure 5.1: System architecture**

# 6. Testing and Result Analysis

## 6.1 Testing Procedure

The testing is a process of checking working of software and hardware products. Testing of software is called software testing and testing of hardware product is called hardware testing. Among these, most commonly used testing are unit testing and system testing. Once the implementation of system has been completed then the entire system is tested to check whether developed system satisfies the requirements specified by the users. The process of testing entire system is called system testing. On the other hand, system is developed by the integration of different modules. Testing of each module of a system is called the unit testing.

## 6.2 Test results

In this work, unit tests are conducted on all modules and obtained the expected results. Later, system testing is conducted on entire system and obtained the expected results. Following table shows unit test cases and results of proposed approach.

**Table 6.1: Unit Test Cases for Proposed Approach**

| Test Case Number | Input | Stage | Expected Behavior | Observed Behavior | Status P=Pass F=Fail |
|---|---|---|---|---|---|
| 1 | User name, Email, Password | Registration | Information Inserted into Database Successfully | As expected | P |
| 2 | User name, Password | Login | Successful Login and redirect to next page | As expected | P |
| 3 | Data Entered | Disease predicted | Displays weather patient has a disease or not | As expected | P |

# 6.3 Result Analysis

Results basically refer to any particular output that comes as a result of the completion of the activities that have been performed as part of the project or a particular project component.

### Home Page

The figure 6.1 tells about home page where user can register or login to the website.

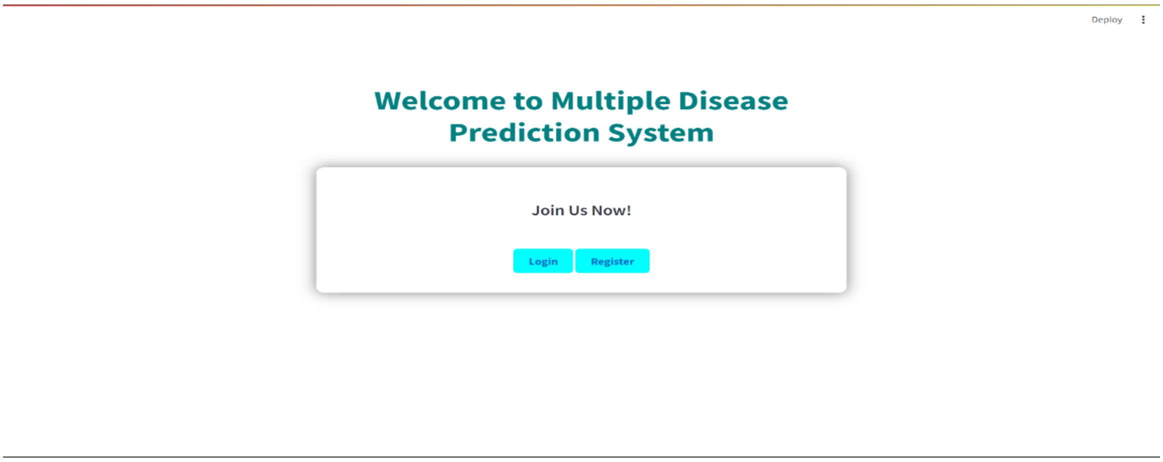


**Figure 6.1: Home Page**

### Register Page

The figure 6.2 is registration page for new user, user can register by giving their username, password and email.

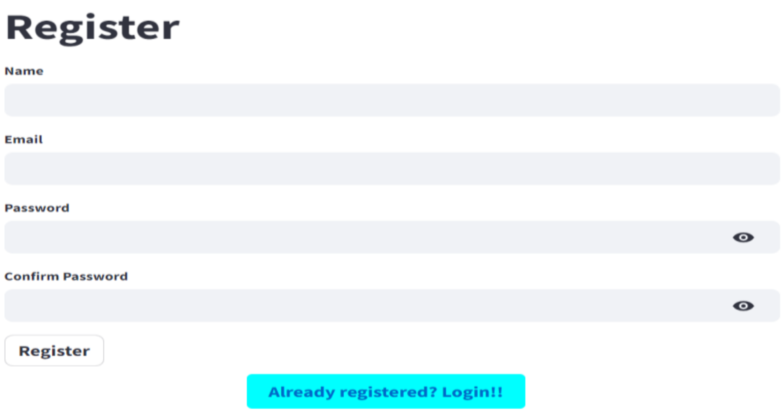


**Figure 6.2: Register page**

## Login Page

The figure 6.3 is the login page where registered users can login using email and password.

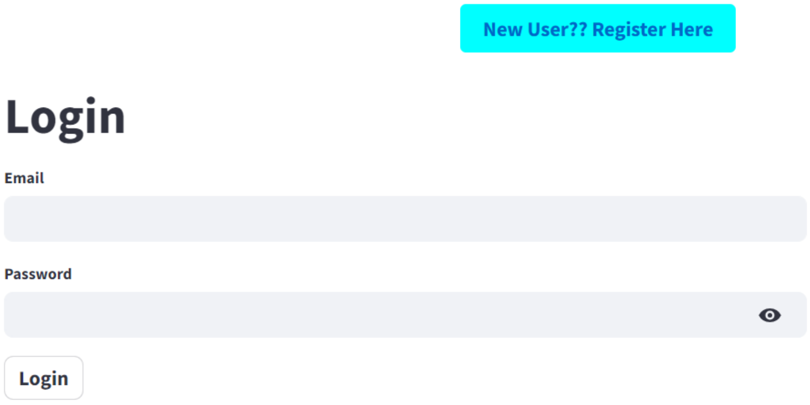


**Figure 6.2: Login Page**

## Diabetes Prediction Page

The figure 6.4 shows the diabetes prediction page where user can enter the specified values and get the result

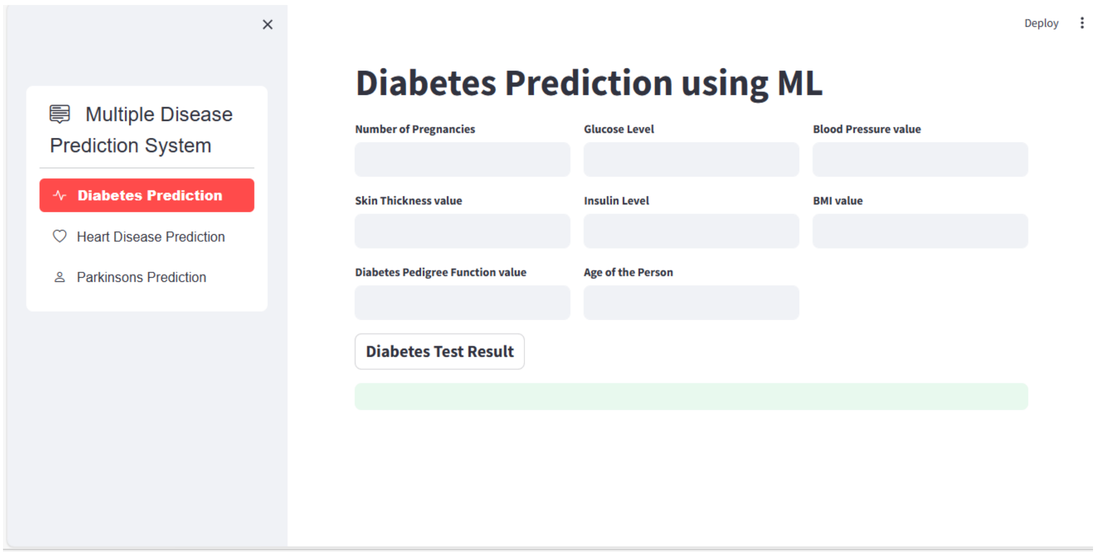


**Figure 6.4 Diabetes Prediction Page**

### Heart Disease Prediction Page

The figure 6.5 is the heart disease prediction page where user can enter the specified values and get the result.

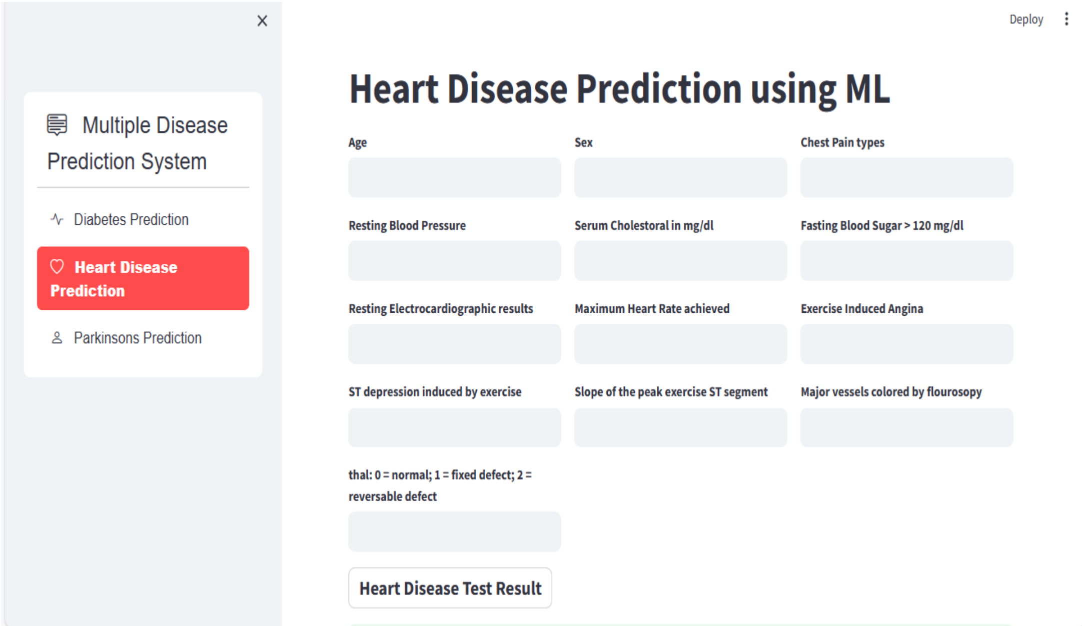


**Figure 6.5: Heart Disease Prediction System**

### Parkinson's Disease Prediction Page

The figure 6.6 is the Parkinson's disease prediction page where user can enter the specified values and get the result.

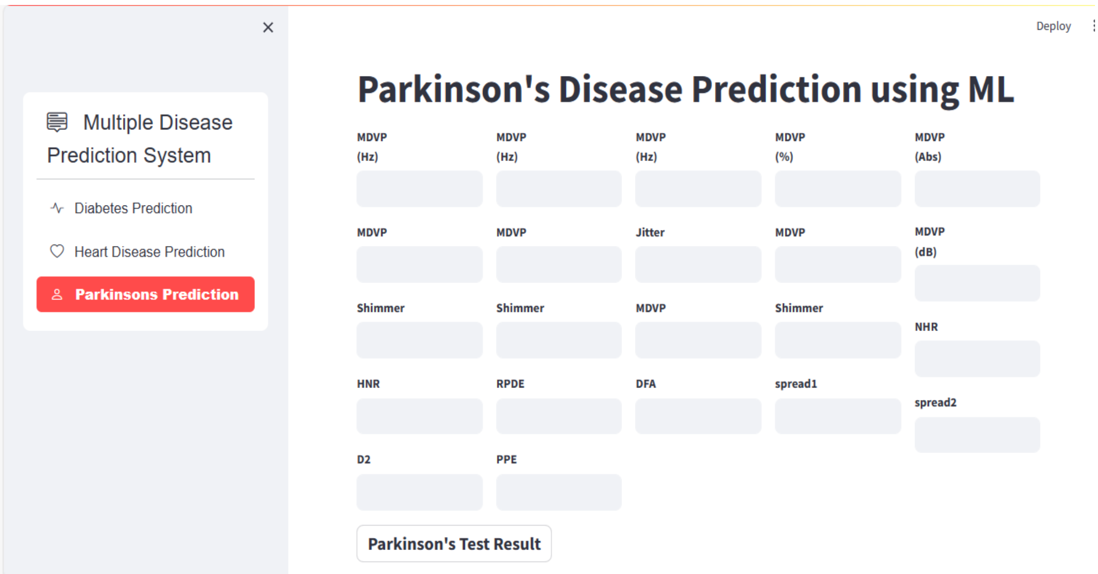

**Figure 6.5: Parkinson's Disease Prediction System**

# 7. Discussion

## 7.1 Common faults observed

An essential element of successful web application is web traffic management, servers and loading time and some specific HTML title tags for each page. During internship project some common faults observed, sometimes forgot to put the title that confuses the user where they are in the web page. Scalability is a difficult thing for web developers to manage. Scalability is load balancing between the servers, hence when the load increases (i.e. more traffic on the page) additional servers can be added to balance it. Slow web applications are a failure and as a result, customers abscond your website, thus damaging your revenue as well as reputation. Security is something the web developers need to consider at every stage of SDLC (software development life cycle).

## 7.2 Causes and Remedies

**Using high speed Internet:**

If the internet speed is slow, it can take longer for the web browser to load these templates onto the page. To address this issue, it is recommended to use a faster internet connection to ensure faster loading of these components.

**Insufficient data:**

One way to address insufficient data is to gather more data. By gathering more data, recommendation becomes more accurate.

**Scalability:**

The entire load should not be thrown on a single server instead of which the software should be designed in such a way that it can work on multiple servers. Service oriented architecture helps developers in managing and improving scalability.

**Poor model selection:**

It is important to choose the most appropriate machine learning algorithm or model architecture for the recommendation task. Experimenting with different models and evaluating their performance can help in selecting the most appropriate one.

# 8. Conclusions

## 8.1 Outcome of the internship

On the whole, this internship was a useful experience. I have gained new knowledge, skills and met many new people. After completing my internship training, I had better understanding of team work. This also helped me to sharpen my skills in Artificial Intelligence and Machine Learning. The internship in Techciti software was a great opportunity for me. The experience I have got is huge and useful. After this internship, I know what I can expect in future career in industry. Skills I learnt will give me the chance of the future personal and professional growth. Advantages and disadvantages of this experience made me stronger in work and more flexible. The result I had got was better than I ever expected. As a personal experience, this internship made my communication skills better and all around. While doing the project I have learned about HTML, CSS, Basics of Bootstrap and Django. The establishment by itself is perfect for internship.

## 8.2 Scope for future work

The future scope is to 3incorporate diverse data sources such as electronic health records (EHRs), genetic data, environmental factors, lifestyle information, wearable sensor data, and social determinants of health. This integration can provide a more comprehensive view for accurate prediction. Explore and develop more sophisticated machine learning algorithms, such as deep learning, ensemble methods, and reinforcement learning, to improve prediction accuracy and handle complex, high-dimensional data. Implement systems for real-time monitoring of health parameters and provide personalized feedback and recommendations to individuals to mitigate disease risks. Develop models that can provide personalized risk assessments for individuals based on their unique characteristics, including genetic predispositions, lifestyle choices, and environmental exposures. Analyze longitudinal data to track disease progression and identify early warning signs of diseases, enabling timely interventions and personalized treatment strategies. Design interpretable machine learning models that provide insights into the features contributing to disease prediction, facilitating better understanding by healthcare professionals and patients. Develop privacy-preserving techniques such as federated learning, differential privacy, and blockchain-based solutions to ensure the security and confidentiality of sensitive health data.

# 9. SWOT Analysis

SWOT analysis is a vital strategic planning tool that can be used by managers to do a situational analysis of the organization. It is an important technique to analyze the present Strengths(S), Weakness (W), Opportunities (O) and Threats (T).

## Strengths

The main strength of this company is its highly skilled and creative workforce through successful training and learning programs. It is investing huge resources in training and development of its employees resulting in a stronger workforce that is not only highly skilled but also motivated to achieve more.

## Weakness

Financial planning is not done properly and efficiently. Company's profile is not much upgraded hence its not known to many and thus clients are less.

## Opportunities

There are job opportunities for freshers after internship training. Profit sharing is given to their employees. Communication skills are improved by their employees by conducting seminars which is held every week.

## Threats

As it is a startup company, they should be more agile because there are a lot of competitions in India among startups. As the company is operating in numerous countries it is exposed to currency fluctuations especially given the volatile political climate in number of markets across the world.

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